

ETA Prediction System

An **ETA(Estimated Time of Arrival) Prediction System** is an advanced digital tool that uses real-time data and artificial intelligence to calculate when a vehicle, shipment, or delivery will reach its destination

Why Simple Rule base ETA fails?

ETA = distance / speed

Problems:

- ignores traffic
- ignores time of day
- ignores road type
- ignores weather
- ignores driver behavior

REAL UBER APPROACH

Uber uses ML model:

ETA = f(distance, traffic, road, driver, time, region)

FEATURE ENGINEERING (MOST IMPORTANT)

We convert ride into features:

distance_km	X = [	Traffic
route_curvature	distance,	Road
number_of_turns Temporal:	hour,	Time
hour_of_day	traffic,	Driver
day_of_week	driver_speed,	Weather
peak_hour_flag Traffic:	road_complexity	Region
congestion_level	]	Route complexity
avg_speed_zone Driver:		Store preparation
driver_speed_avg		Historical behavior
driver_history_delay Context:		
rain / weather		
event zone		

SIMPLE WORKING MODEL (JAVA SIMULATION)

```
public class EtaPredictionService {
    public double predictETA(
        double distanceKm,
        int hourOfDay,
        double trafficLevel,
        double driverSpeed,
        double roadComplexity
    ){
        // Simulated ML weights (like regression model)
        double eta =
            (distanceKm * 2.5) +
            (trafficLevel * 8) +
            (roadComplexity * 3) -
            (driverSpeed * 1.2) +
            (isPeakHour(hourOfDay) ? 5 : 0);

        return Math.max(1, eta); // minutes
    }

    private boolean isPeakHour(int hour) {
        return (hour >= 8 && hour <= 11) ||
            (hour >= 17 && hour <= 21);
    }
}
```

REAL CALL FLOW:

Ride Request

↓

Feature Builder Service

↓

ML Model Server (Python)

↓

ETA Prediction

↓

Driver Matching + Pricing

For a delivery platform, ETA can mean:

Order placed → Store preparation → Driver reaches store →
Pickup → Travel → Customer arrival

Therefore:

Total ETA =

Preparation Time
+ Driver Arrival Time
+ Pickup Time
+ Travel Time
+ Delivery Buffer

For example:

Preparation	= 15 min
Driver arrival	= 5 min
Pickup	= 3 min
Travel	= 18 min
Buffer	= 4 min

Total ETA	= 45 min

1. Basic Rule-Based ETA

The simplest ETA calculation is:

$$\text{ETA} = \text{distance} / \text{speed}$$

This works for a very simple system. But real roads are not simple.

1.1. Why Rule-Based ETA Fails?

Scenario A:

Distance = 5 km

Time = 3 AM

Traffic = Low

ETA = 10 minutes

Scenario B:

Distance = 5 km

Time = 8:30 AM

Traffic = High

ETA = 25 minutes

Same distance. but, Completely different ETA.

1.2. The Problem With Distance / Speed

The distance is identical. But the travel time is different.

$$\text{ETA} = \text{distance} / \text{speed}$$

2. ETA vs Route Optimization

Route optimization answers:

In what order should we visit multiple stops?

Example:

Driver → Store A → Store B → Customer C → Customer D

ETA prediction answers::

How long will this route take?

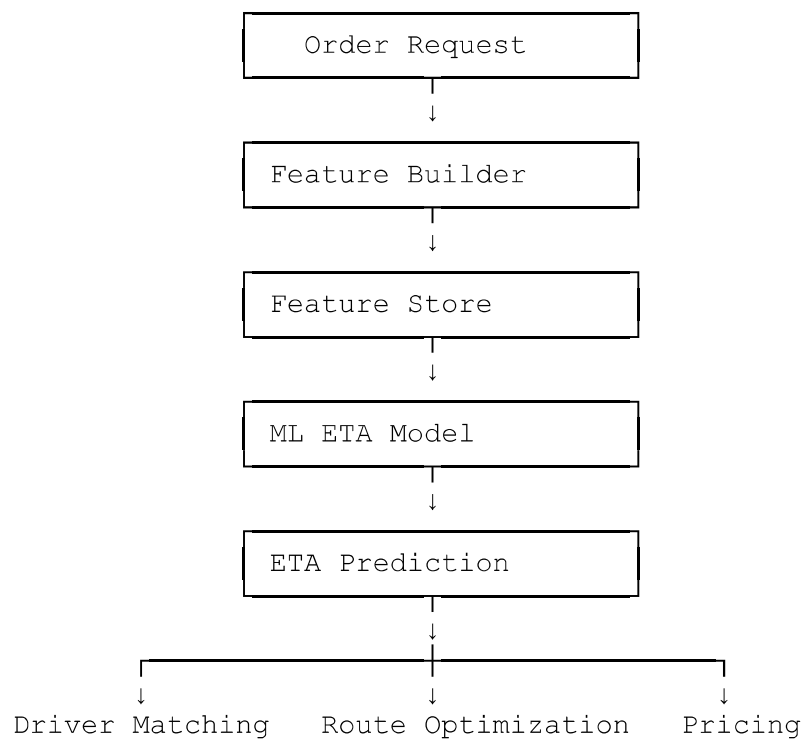
Therefore::

Route Optimization → Best stop sequence → ETA Prediction → Travel time

They work together.

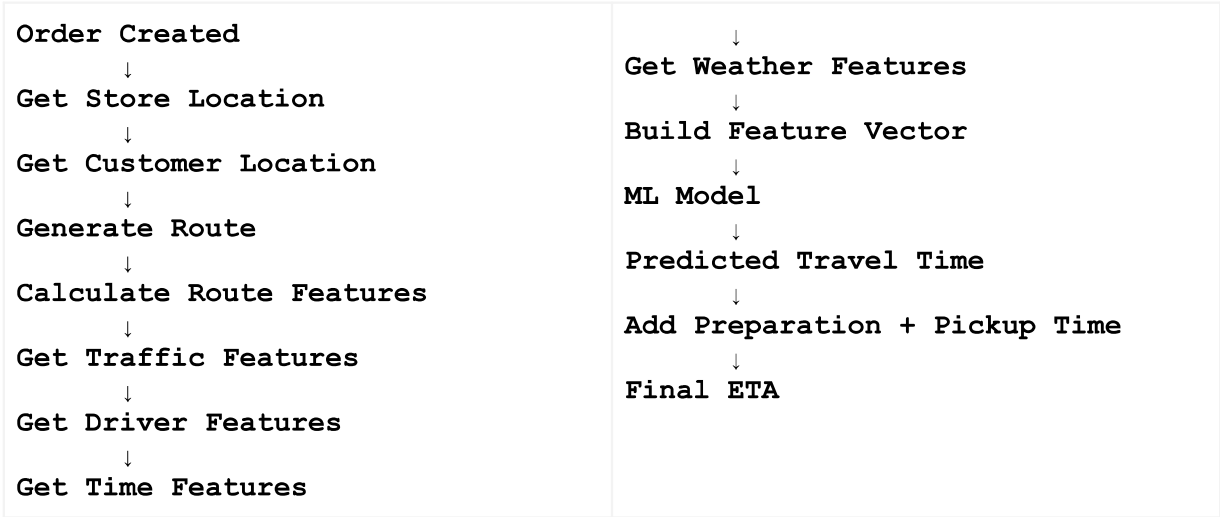
3. ETA Prediction Architecture

A production-style architecture can look like:



4. Complete ETA Flow

The complete flow:



5. ETA Input Data

The ML model needs useful information. We divide features into categories:

Spatial Temporal Traffic Driver Weather Region Order Route Store	Spatial Features: distance_km route_curvature number_of_turns number_of_intersections road_length ----- distance_km = 7.5 number_of_turns = 14 route_curvature = 0.42	Temporal Features: hour_of_day day_of_week month weekend_flag holiday_flag peak_hour_flag ----- hour_of_day = 18 day_of_week = 5 peak_hour_flag = true
---	---	---

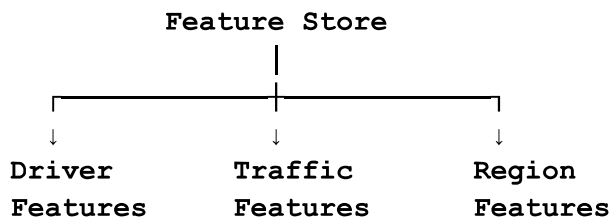
Traffic Features: congestion_level average_speed current_speed historical_speed traffic_incidents road_blockage ----- congestion_level = 0.82 avg_speed_zone = 18 km/h	Driver Features: driver_speed_avg driver_history_delay driver_delivery_count driver_avg_stop_time driver_route_adherence ----- Driver A Average speed = 28 km/h Driver B Average speed = 21 km/h	Weather Features: rain temperature visibility weather_severity ----- rain = true weather_severity = 0.7 Heavy rain can increase: Traffic Travel time Pickup delay Delivery delay
--	---	---

Region Features: city zone polygon_id neighborhood road_density traffic_density ----- polygon_id = HYD_042	Order Features: order_type number_of_items order_value special_instructions pickup_complexity delivery_type ----- Large or complex orders may increase preparation or pickup time.	Route Complexity Features: number_of_turns number_of_signals number_of_intersections highway_ratio local_road_ratio road_complexity ----- Route A: 5 km 4 turns Route B: 5 km 17 turns Route B may take significantly longer.
---	---	--

6. Feature Store

A production system can maintain frequently used features in a Feature Store.

Conceptually:



Example:

driver_speed_avg
driver_delay_rate
zone_avg_speed
zone_congestion
region_delivery_delay

7. Feature Vector

All features are combined into a feature vector:

X = [distance, hour, day_of_week, traffic, avg_speed, driver_speed, driver_delay, road_complexity, turns, weather, region]	Example Feature Vector: distance_km = 8.2 hour_of_day = 18 day_of_week = 5 traffic_level = 0.75 avg_speed_zone = 19 driver_speed_avg = 24 driver_delay = 2.1 road_complexity = 0.62 number_of_turns = 12 rain = 1 region_id = 42	X = [8.2, 18, 5, 0.75, 19, 24, 2.1, 0.62, 12, 1, 42]
--	--	--

8. Feature Normalization

All features are combined into a feature vector:

normalized_x = (x - mean) / standard_deviation

9. Historical Data

Machine Learning needs historical data. For every completed delivery, we can store:

Order ID	Traffic	Predicted ETA
Driver ID	Weather	Actual Travel
Route	Time	Time
Distance	Features	

10. ETA Training Dataset

A simplified training dataset:

Distance	Traffic	Hour	Driver Speed	Turns	Weather	Actual ETA
5.2	0.2	11	28	5	0	13
7.4	0.7	18	22	11	0	28
4.1	0.8	19	20	8	1	24
9.2	0.4	15	26	14	0	15

11. Label Generation

The target variable is usually:

actual_travel_time

For example:

Prediction time = 10:00

Arrival time = 10:27

Actual travel time = 27 minutes

Training record:

X = route/context features

Y = 27

12. Model Options

Possible models:

Linear Regression

Random Forest

Gradient Boosting

XGBoost ★

LightGBM

Neural Networks

Deep Spatio-Temporal Models

Linear Regression

ETA =

$w_1 * \text{distance}$

$+ w_2 * \text{traffic}$

$+ w_3 * \text{turns}$

$+ w_4 * \text{driver_speed}$

$+ \text{bias}$

ETA Model Input:

13. ETA Model Input/Output

Example ETA Prediction

Request:

```
{
  "distanceKm": 8.2,
  "hourOfDay": 18,
  "dayOfWeek": 5,
  "trafficLevel": 0.75,
  "avgSpeedZone": 19,
  "driverSpeedAvg": 24,
  "driverHistoryDelay": 2.1,
  "roadComplexity": 0.62,
  "numberOfTurns": 12,
  "rain": true
}
```

ETA Prediction Response:

```
{
  "etaMinutes": 27.4,
  "confidence": 0.87,
  "modelVersion": "eta-xgb-v12"
}
Or:
{
  "predictedTravelTimeMinutes": 27.4,
  "lowerBoundMinutes": 24,
  "upperBoundMinutes": 32
}
```

14. Simple Java Simulation

```
@Service
public class EtaPredictionService {
    public double predictETA(
        double distanceKm,
        int hourOfDay,
        double trafficLevel,
        double driverSpeed,
        double roadComplexity
    ){
        // Simulated ML weights
        double eta =
            (distanceKm * 2.5) +
            (trafficLevel * 8) +
            (roadComplexity * 3) -
            (driverSpeed * 1.2) +
            (isPeakHour(hourOfDay) ? 5 : 0);

        return Math.max(1, eta);
    }

    private boolean isPeakHour(int hour) {
        return (hour >= 8 && hour <= 11) || (hour >= 17 && hour <= 21);
    }
}
```

15. ETA Prediction Service

```
@Service
public class EtaPredictionService {
    public EtaPredictionResponse predict(EtaPredictionRequest request) {
        double eta = calculatePrediction(request);
        return new EtaPredictionResponse(
            eta,
            "SIMULATED_MODEL_V1"
        );
    }
}
```

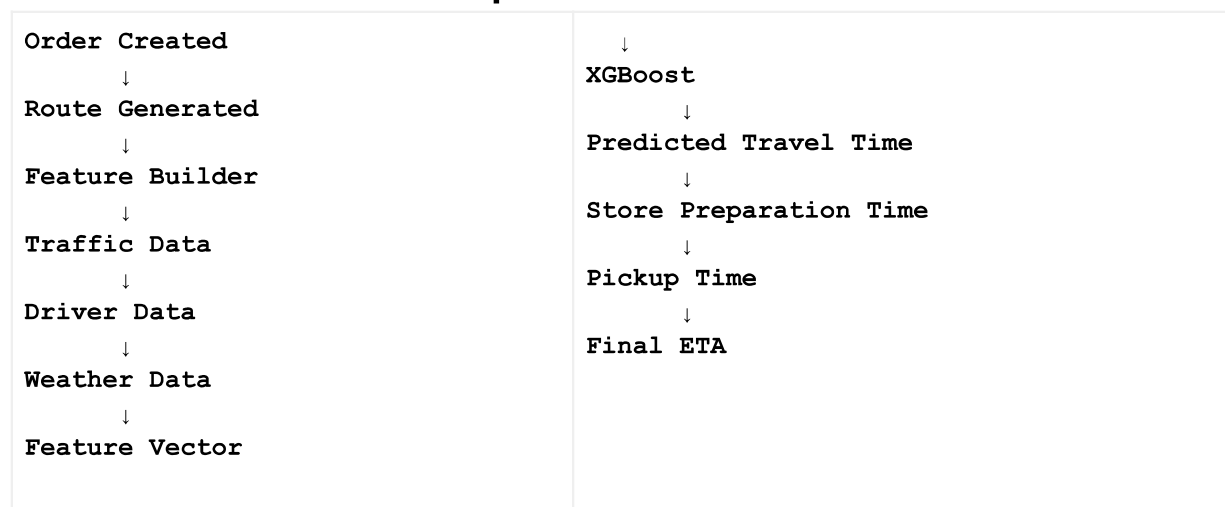
This gives us a stable application-level interface. Later, we can replace the implementation with a real ML model.

16. Feature Builder Service

The Feature Builder combines data from different systems.

```
public EtaFeatures build(Order order, Driver driver, Route route, TrafficData traffic) {
    // build model features
}
```

17. Complete Prediction Flow



18. Driver-Specific ETA

A production system should ideally consider the assigned driver's behavior.

Example:

Driver A Historical average speed = 28 km/h Driver B Historical average speed = 21 km/h	Same route: Driver A ETA = 18 min Driver B ETA = 24 min
--	--

This allows the model to personalize predictions.

19. Traffic-Aware ETA

Traffic should be represented using current and historical information.

Current traffic = 0.85 Historical traffic at 8 PM = 0.72	The model can combine both: Current condition + Historical pattern ↓ ETA
---	--

20. Time-of-Day ETA

Traffic varies throughout the day.

02:00 → Low 07:00 → Increasing 09:00 → High 14:00 → Medium 18:00 → Very High 23:00 → Low	Features: hour_of_day day_of_week peak_hour weekend holiday
---	---

21. Weather-Aware ETA

Traffic varies throughout the day.

Normal weather:

ETA = 20 min

Heavy rain:

ETA = 28 min

The model can learn this relationship from historical deliveries.

22. Road-Type ETA

Road types have different characteristics.

Highway

Main road

Residential road

Narrow road

Service road

Feature:

highway_ratio

local_road_ratio

road_complexity

average_speed

23. Region-Aware ETA

Different regions can have different travel patterns.

Region A: High traffic
Low average speed

Region B: Low traffic
High average speed

A region feature helps the model learn these differences.

24. Store Preparation Time

Supposr:

Travel = 20 min | Store preparation = 15 min

Then: Customer ETA $\neq$ 20 min

Instead: $ETA = 15 + 20 = 35$ min

Store preparation can itself become an ML prediction problem.

25. Pickup ETA

Pickup ETA answers:

When will the driver reach the store?

Example:

Driver $\rightarrow$ Store

Model:

Distance
Traffic
Driver
Road
Time

Output:

Driver arrival = 8 minutes

26. Delivery ETA

Delivery ETA answers:

When will the customer receive the order?

Example:

Store $\rightarrow$ Driver $\rightarrow$ Customer

It includes::

Store preparation + Driver arrival +
Pickup + Travel + Delivery

27. Multi-Stop ETA

Driver $\rightarrow$ Store A $\rightarrow$ Customer A $\rightarrow$
Store B $\rightarrow$ Customer B

ETA must consider all stops.

$ETA = \text{Travel A} + \text{Pickup A} + \text{Delivery A} +$
 $\text{Travel B} + \text{Pickup B} + \text{Delivery B}$

This is where ETA Prediction and Route Optimization become tightly connected

28. ETA for Multi-Drop Routes

The route optimizer provides:

Stop sequence

ETA system predicts:

Time between stops

Architecture:

Route Optimizer



Stop Sequence



ETA Model



Travel Time Per Segment



Total Route ETA

29. ETA Confidence

A good ETA system should not always return only one number.

Example:

ETA = 27 minutes

Confidence = 87%

Confidence can be derived from:

Model uncertainty

Historical prediction error

Traffic stability

Data quality

Route complexity

30. ETA Range

A more useful customer-facing prediction can be:

Delivery in 25–32 minutes

instead of:

Delivery in exactly 27 minutes

The backend can maintain:

Predicted ETA = 27

Lower = 25

Upper = 32

31. ETA Error

After the delivery completes:

Predicted ETA = 27 min

Actual ETA = 31 min

Error: $31 - 27 = +4$ minutes

This error becomes important training and monitoring data.

32. Measuring Model Accuracy

MAE

RMSE

MAPE

Median Absolute Error

P90 Error

P95 Error

32.1 MAE

Mean Absolute Error:

```
MAE =  
average(|actual - predicted|)
```

On average, the prediction is off by 2.5 minutes.

32.2 RMSE

RMSE penalizes larger errors more strongly.

```
RMSE =  
sqrt(  
  average(  
    (actual - predicted)^2  
  )  
)
```

Useful when large ETA errors are particularly costly.

32.3 MAPE

Mean Absolute Error:

```
MAPE =  
average(  
  |actual - predicted| / actual  
)  
× 100
```

It expresses error as a percentage. However, MAPE can behave poorly when actual values are very small, so it should not be used blindly.

32.4 Prediction Error Monitoring

Monitor::

```
Average error  
P50 error  
P90 error  
P95 error  
Overprediction rate  
Underprediction rate
```

33. ETA Recalculation

Recalculate ETA when important state changes occur.

Example:

Driver location changed
Traffic changed
Route changed
Store preparation changed
Weather changed
New stop added
Stop completed

Do not necessarily recalculate on every GPS packet.

Use controlled intervals or significant state changes to avoid unnecessary model calls.

34. Operations Dashboard

Order	ETA	Traffic	Risk
1001	24m	Low	Low
1002	31m	High	Medium
1003	42m	Severe	High